1 Discrete Joint Probability Distribution

Let the joint probability distribution be defined as:

$$P(X = x, Y = y) = \begin{cases} \frac{1}{10}, & \text{for } (x, y) \in \{(0,0), (0,1), (1,0), (1,1), (2,1)\} \\ 0, & \text{otherwise} \end{cases}$$

$X \backslash Y$	0	1	$P_X(x)$
0	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{2}{10}$
1	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{2}{10}$
2	0	$\frac{1}{10}$	$\frac{1}{10}$
$P_Y(y)$	$\frac{2}{10}$	$\frac{3}{10}$	1

(a) List all possible values of the random variables X and Y.

(b) Find the marginal probability distribution of X.

(c) Find the marginal probability distribution of Y.

(d) Find the joint probability $P(X = 1, Y = 1)$.

(e) Find the probability $P(X = 2)$.

(f) Find the probability $P(Y = 0)$.

(g) Compute $P(X = 0 \text{ or } Y = 1)$.

(h) Compute $P(X \leq 1, Y = 1)$.

(i) Find the conditional probability $P(Y = 1 \mid X = 0)$.

(j) Find the conditional probability $P(X = 2 \mid Y = 1)$.

(k) Find the conditional distribution of Y given $X = 1$.

(l) Find the conditional distribution of X given $Y = 0$.

(m) Are X and Y independent random variables? Justify your answer.

(n) Verify independence by checking whether $P(X = 1, Y = 1) = P_X(1) \cdot P_Y(1)$.

(o) Explain whether knowing the value of X affects the distribution of Y.

(p) Find the expected value $E[X]$.

(q) Find the expected value $E[Y]$.

(r) Find the expected value $E[XY]$.

(s) Compute the variance $\text{Var}(X)$.

(t) Compute the variance $\text{Var}(Y)$.

(u) Compute the standard deviations of X and Y.

(v) Find the covariance $\text{Cov}(X, Y)$.

(w) Find the correlation coefficient $\rho_{X,Y}$.

(x) Interpret the sign and magnitude of the correlation.

(y) Determine if there is a linear relationship between X and Y.

2 Continuous Joint Probability Distribution

A chemical system that results from a chemical reaction has two important components among others in a blend. The joint probability density function (pdf) describing the proportions X_1 and X_2 of these two components is given by:

$$f(x_1, x_2) = \begin{cases} 2, & 0 < x_1 < x_2 < 1 \\ 0, & \text{otherwise.} \end{cases}$$

(a) Sketch the region over which the joint pdf is defined.

(b) Verify that $f(x_1, x_2)$ is a valid joint probability density function.

(c) Find the marginal probability density function of X_1.

(d) Find the marginal probability density function of X_2.

(e) Find $P(X_1 < 0.2, X_2 > 0.5)$.

(f) Find the conditional pdf $f_{X_1|X_2}(x_1|x_2)$.

(g) Find the conditional pdf $f_{X_2|X_1}(x_2|x_1)$.

(h) Compute the conditional probability $P(X_1 < 0.5 \mid X_2 = 0.8)$.

(i) Are X_1 and X_2 independent? Justify your answer.

(j) Compute the expected value $E[X_1]$.

(k) Compute the expected value $E[X_2]$.

(l) Compute the expected value $E[X_1 X_2]$.

(m) Compute $\text{Var}(X_1)$ and $\text{Var}(X_2)$.

(n) Find Median

(o) Find the covariance $\text{Cov}(X_1, X_2)$.

(p) Compute the correlation coefficient ρ_{X_1,X_2} and interpret its meaning.

(q) Explain whether X_1 and X_2 exhibit a linear relationship based on correlation.

3 Bayes' Rule

If A and B are two events such that $P(B) > 0$, then Bayes' theorem is given by:

$$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

- $P(A)$: Prior probability of event A
- $P(B \mid A)$: Likelihood of event B given A
- $P(B)$: Total (marginal) probability of event B
- $P(A \mid B)$: Posterior probability of event A given B

3.1 Generalized Bayes' Rule (Using Total Probability)

Let $\{A_1, A_2, \ldots, A_n\}$ be a partition of the sample space such that $P(A_i) > 0$ for all i, and let B be any event with $P(B) > 0$. Then, for any $k = 1, 2, \ldots, n$:

$$P(A_k \mid B) = \frac{P(B \mid A_k) \cdot P(A_k)}{\sum_{i=1}^{n} P(B \mid A_i) \cdot P(A_i)}$$

This is especially useful when B can occur due to multiple causes $A_1, A_2, \ldots, A_n$, and we are interested in finding the probability of a specific cause given that B has occurred.

3.2 Questions

(a) A medical test for a disease is 99% accurate for diseased individuals and 95% accurate for healthy individuals. The disease occurs in 1% of the population. What is the probability that a person has the disease given that they tested positive?

(b) In a factory, 40% of products are made by Machine A and 60% by Machine B. Machine A produces 5% defective items, and Machine B produces 10% defective items. If an item is found to be defective, what is the probability it was produced by Machine A?

(c) In a certain city, 70% of people prefer tea, while 30% prefer coffee. Among tea lovers, 60% like it with milk. Among coffee lovers, only 20% like it with milk. A randomly chosen person likes milk in their drink. What is the probability they prefer tea?

(d) A bag contains 3 red balls and 2 blue balls (Bag A). Another bag (Bag B) contains 4 red balls and 5 blue balls. A bag is chosen at random, and a red ball is drawn. What is the probability that Bag A was chosen?

(e) A diagnostic test gives a positive result 90% of the time when a person has a condition, and 8% of the time when they don't. If 5% of the population has the condition, what is the probability a person with a positive test result actually has the condition?

(f) A student is trying to determine which of three professors (A, B, C) graded his exam. Prof A gives an average grade of 85 with 70% probability, Prof B gives an average of 75 with 20%, and Prof C gives an average of 95 with 10%. The student receives a grade of 95. What is the probability Prof C graded the exam?

(g) A company has three warehouses supplying parts: Warehouse 1 supplies 50% of parts and has 2% defect rate, Warehouse 2 supplies 30% with 3% defect rate, and Warehouse 3 supplies 20% with 5% defect rate. A defective part is found. What is the probability it came from Warehouse 3?

(h) A school has 60% day students and 40% boarding students. 90% of day students pass the final exam, and 80% of boarding students pass. If a student is known to have passed, what is the probability that they are a boarding student?

4 Continuous Probability Distributions

4.1 Summary

Distribution	PDF $f(x)$	Domain	MGF $M_X(t)$	Mean, Variance
Uniform $U(a,b)$	$\frac{1}{b-a}$	$a < x < b$	$\frac{e^{tb}-e^{ta}}{t(b-a)},\ t \neq 0$	$\mu = \frac{a+b}{2}$ $\sigma^2 = \frac{(b-a)^2}{12}$
Exponential $\text{Exp}(\lambda)$	$\lambda e^{-\lambda x}$	$x \geq 0$	$\frac{\lambda}{\lambda - t},\ t < \lambda$	$\mu = \frac{1}{\lambda}$ $\sigma^2 = \frac{1}{\lambda^2}$
Gamma $\text{Gamma}(\alpha,\lambda)$	$\frac{\lambda^\alpha x^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)}$	$x > 0$	$\left(\frac{\lambda}{\lambda - t}\right)^\alpha,\ t < \lambda$	$\mu = \frac{\alpha}{\lambda}$ $\sigma^2 = \frac{\alpha}{\lambda^2}$
Normal $N(\mu,\sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	$-\infty < x < \infty$	$e^{\mu t + \frac{1}{2}\sigma^2 t^2}$	$\mu = \mu$ $\sigma^2 = \sigma^2$

4.2 The Uniform Distribution

Let $X \sim U(a,b)$. The probability density function (pdf) is:

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$$

4.2.1 Proof of Valid PDF

To confirm $f(x)$ is a valid PDF, two conditions must hold:

1. $f(x) \geq 0$ for all x.
Since $\frac{1}{b-a} > 0$, and outside $[a,b]$ it is zero, this condition is satisfied.
2. The total integral over the domain equals 1:

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_a^b \frac{1}{b-a}\,dx = \frac{1}{b-a}\int_a^b dx = \frac{1}{b-a}(b-a) = 1$$

Thus, the uniform distribution on $[a,b]$ is a valid probability density function.

4.2.2 Moment Generating Function (MGF)

The moment generating function is defined as:

$$M_X(t) = \mathbb{E}[e^{tX}] = \int_a^b e^{tx} \cdot \frac{1}{b-a}\,dx = \frac{1}{b-a}\int_a^b e^{tx}\,dx$$

$$= \frac{1}{b-a} \cdot \left[\frac{e^{tx}}{t}\right]_a^b = \frac{1}{t(b-a)}\left(e^{tb} - e^{ta}\right), \quad t \neq 0$$

$$\boxed{M_X(t) = \frac{e^{tb}-e^{ta}}{t(b-a)}, \quad t \neq 0}$$

4.2.3 Mean $\mathbb{E}[X]$ using L'Hôpital's Rule

$$\mathbb{E}[X] = M_X'(0) = \lim_{t\to 0} \frac{d}{dt}\left(\frac{e^{tb}-e^{ta}}{t(b-a)}\right)$$

Let:

$$g(t) = e^{tb} - e^{ta}, \quad h(t) = t(b-a)$$

Then:

$$M_X'(0) = \lim_{t\to 0} \frac{g'(t)}{h'(t)} = \left.\frac{be^{tb}-ae^{ta}}{b-a}\right|_{t=0} = \frac{b-a}{b-a} = 1$$

However, recall that $M_X(t)$ includes a factor $\frac{1}{b-a}$, so the correct expression is:

$$M_X(t) = \frac{1}{b-a} \cdot \frac{e^{tb}-e^{ta}}{t}$$

Now apply L'Hôpital's Rule directly:

$$M_X'(0) = \frac{1}{b-a} \cdot \lim_{t\to 0} \frac{be^{tb}-ae^{ta}}{1} = \frac{b-a}{b-a} \cdot \frac{a+b}{2} = \boxed{\frac{a+b}{2}}$$

4.2.4 Second Moment $\mathbb{E}[X^2]$ using L'Hôpital's Rule

Start from the first derivative of $M_X(t)$:

$$M_X'(t) = \frac{(be^{tb}-ae^{ta})t - (e^{tb}-e^{ta})}{t^2(b-a)}$$

Now compute the second derivative:

$$M_X''(0) = \lim_{t\to 0} \frac{t(b^2e^{tb}-a^2e^{ta})}{2t(b-a)} = \frac{b^2-a^2}{2(b-a)} = \frac{a+b}{2}$$

Wait — that is incorrect for $\mathbb{E}[X^2]$; let's proceed carefully.
Instead, differentiate numerator and denominator separately using L'Hôpital's Rule again:
Numerator:

$$N(t) = (be^{tb}-ae^{ta})t - (e^{tb}-e^{ta}) \Rightarrow N'(t) = (be^{tb}-ae^{ta}) + t(b^2e^{tb}-a^2e^{ta}) - (be^{tb}-ae^{ta}) = t(b^2e^{tb}-a^2e^{ta})$$

Denominator:

$$D(t) = t^2(b-a) \Rightarrow D'(t) = 2t(b-a)$$

So:

$$M_X''(0) = \lim_{t\to 0} \frac{t(b^2e^{tb}-a^2e^{ta})}{2t(b-a)} = \frac{b^2-a^2}{2(b-a)} = \boxed{\frac{a^2+ab+b^2}{3}}$$

4.2.5 Variance

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{a^2+ab+b^2}{3} - \left(\frac{a+b}{2}\right)^2 = \frac{(b-a)^2}{12}$$

4.2.6 Questions

(a) Find the mean and variance of a random variable X that is uniformly distributed on the interval $[2, 10]$.

(b) If $X \sim \text{Uniform}(0, 5)$, find $P(1 < X < 3)$.

(c) Suppose $X \sim \text{Uniform}(a, b)$ and you are given that $\mathbb{E}[X] = 7$ and $\mathrm{Var}(X) = 4$. Find the values of a and b.

(d) Let $X \sim \text{Uniform}(-3, 3)$. Compute $\mathbb{E}[X^3]$ and $\mathrm{Var}(X^2)$.

4.3 The Exponential Distribution

PDF of the Exponential Distribution

Let $X \sim \text{Exponential}(\lambda)$, where $\lambda > 0$. Then the probability density function (pdf) is:

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

4.3.1 Proof of Valid PDF

To confirm $f(x)$ is a valid PDF, two conditions must hold:

1. $f(x) \geq 0$ for all x. This is true since $\lambda > 0$ and $e^{-\lambda x} > 0$ for $x \geq 0$.
2. The total integral over the domain equals 1:

$$\int_{-\infty}^{\infty} f(x)\, dx = \int_{0}^{\infty} \lambda e^{-\lambda x} dx = 1$$

Compute the integral:

$$\int_{0}^{\infty} \lambda e^{-\lambda x} dx = \lambda \int_{0}^{\infty} e^{-\lambda x} dx$$

Recall that

$$\int_{0}^{\infty} e^{-\lambda x} dx = \left[-\frac{1}{\lambda} e^{-\lambda x} \right]_{0}^{\infty} = 0 + \frac{1}{\lambda} = \frac{1}{\lambda}$$

Therefore,

$$\int_{0}^{\infty} \lambda e^{-\lambda x} dx = \lambda \cdot \frac{1}{\lambda} = 1$$

4.3.2 Moment Generating Function (MGF)

The moment generating function is defined by:

$$M_X(t) = \mathbb{E}[e^{tX}] = \int_{0}^{\infty} e^{tx} \cdot \lambda e^{-\lambda x}\, dx = \lambda \int_{0}^{\infty} e^{-(\lambda - t)x}\, dx$$

This integral converges for $t < \lambda$. Then:

$$M_X(t) = \lambda \cdot \left[\frac{e^{-(\lambda - t)x}}{-(\lambda - t)} \right]_{0}^{\infty} = \lambda \cdot \left(0 - \frac{1}{-(\lambda - t)} \right) = \frac{\lambda}{\lambda - t}, \quad t < \lambda$$

$$\boxed{M_X(t) = \frac{\lambda}{\lambda - t}, \quad \text{for } t < \lambda}$$

4.3.3 First Moment $\mathbb{E}[X]$

Differentiate $M_X(t)$:

$$M_X'(t) = \frac{d}{dt} \left(\frac{\lambda}{\lambda - t} \right) = \frac{\lambda}{(\lambda - t)^2}$$

Then evaluate at $t = 0$:

$$\mathbb{E}[X] = M_X'(0) = \frac{\lambda}{\lambda^2} = \boxed{\frac{1}{\lambda}}$$

4.3.4 Second Moment $\mathbb{E}[X^2]$

Differentiate again to get the second derivative:

$$M_X''(t) = \frac{d}{dt}\left(\frac{\lambda}{(\lambda - t)^2}\right) = \frac{2\lambda}{(\lambda - t)^3}$$

Evaluate at $t = 0$:

$$\mathbb{E}[X^2] = M_X''(0) = \frac{2\lambda}{\lambda^3} = \boxed{\frac{2}{\lambda^2}}$$

4.3.5 Variance

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \boxed{\frac{1}{\lambda^2}}$$

4.3.6 Questions

(a) Let $X \sim \text{Exponential}(\lambda = 2)$. Find the mean and variance of X.

(b) For $X \sim \text{Exponential}(\lambda = 3)$, calculate $P(X > 2)$.

(c) Suppose the lifetime of a device follows an exponential distribution with unknown rate λ. Given that the probability the device lasts more than 5 hours is 0.1, find λ.

(d) If $X \sim \text{Exponential}(\lambda = 4)$, compute $P(1 < X < 3)$.

(e) Let $X \sim \text{Exponential}(\lambda = 5)$. Find $\mathbb{E}[X^2]$ using the moment generating function.

4.4 The Gamma Distribution

PDF of the Gamma Distribution

Let $X \sim \text{Gamma}(\alpha, \lambda)$, where $\alpha > 0$ is the shape parameter, and $\lambda > 0$ is the rate parameter.
The probability density function (pdf) is:

$$f(x) = \begin{cases} \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

4.4.1 Proof of Valid PDF

To verify that this is a valid PDF, we must prove two things:

1. $f(x) \geq 0$ for all $x \in \mathbb{R}$
2. $\int_{-\infty}^{\infty} f(x)\, dx = 1$

1. Non-Negativity
For $x > 0$, all components of the PDF are positive:

$$\lambda^\alpha > 0, \quad x^{\alpha-1} \geq 0, \quad e^{-\lambda x} > 0, \quad \Gamma(\alpha) > 0$$

Thus, $f(x) \geq 0$ for all $x \in \mathbb{R}$.
2. Total Area Under the Curve
We evaluate the total integral:

$$\int_{-\infty}^{\infty} f(x)\, dx = \int_0^{\infty} \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}\, dx$$

Use substitution: let $u = \lambda x \Rightarrow x = \frac{u}{\lambda}$, $dx = \frac{du}{\lambda}$

$$\int_0^\infty \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}\, dx = \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty \left(\frac{u}{\lambda}\right)^{\alpha-1} e^{-u} \cdot \frac{1}{\lambda}\, du$$

$$= \frac{\lambda^\alpha}{\Gamma(\alpha)} \cdot \frac{1}{\lambda^\alpha} \int_0^\infty u^{\alpha-1} e^{-u}\, du = \frac{\lambda^\alpha}{\Gamma(\alpha)} \cdot \frac{1}{\lambda^\alpha} \cdot \Gamma(\alpha) = 1$$

4.4.2 Moment Generating Function (MGF)

The moment generating function is defined as:

$$M_X(t) = \mathbb{E}[e^{tX}] = \int_0^\infty e^{tx} \cdot \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}\, dx = \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{\alpha-1} e^{-(\lambda-t)x}\, dx$$

This is the definition of the gamma integral:

$$\int_0^\infty x^{\alpha-1} e^{-\beta x}\, dx = \frac{\Gamma(\alpha)}{\beta^\alpha}, \quad \text{for } \beta > 0$$

So for $t < \lambda$:

$$M_X(t) = \frac{\lambda^\alpha}{\Gamma(\alpha)} \cdot \frac{\Gamma(\alpha)}{(\lambda-t)^\alpha} = \left(\frac{\lambda}{\lambda-t}\right)^\alpha, \quad t < \lambda$$

$$\boxed{M_X(t) = \left(\frac{\lambda}{\lambda-t}\right)^\alpha, \quad t < \lambda}$$

4.4.3 Mean $\mathbb{E}[X]$

We compute the first derivative:

$$M_X'(t) = \frac{d}{dt}\left(\frac{\lambda}{\lambda-t}\right)^\alpha = \alpha \left(\frac{\lambda}{\lambda-t}\right)^\alpha \cdot \frac{1}{\lambda-t}$$

Evaluating at $t = 0$:

$$\mathbb{E}[X] = M_X'(0) = \alpha \left(\frac{\lambda}{\lambda}\right)^\alpha \cdot \frac{1}{\lambda} = \boxed{\frac{\alpha}{\lambda}}$$

4.4.4 Second Moment $\mathbb{E}[X^2]$

Differentiate $M_X'(t)$ again:
Let $M_X(t) = (\lambda/(\lambda-t))^\alpha$. Then:

$$M_X''(t) = \frac{d}{dt}\left[\alpha \left(\frac{\lambda}{\lambda-t}\right)^\alpha \cdot \frac{1}{\lambda-t}\right]$$

Using the product rule:

$$M_X''(t) = \alpha(\alpha+1)\left(\frac{\lambda}{\lambda-t}\right)^{\alpha+1} \cdot \frac{1}{(\lambda-t)^2}$$

Now evaluate at $t = 0$:

$$\mathbb{E}[X^2] = M_X''(0) = \alpha(\alpha+1)\left(\frac{\lambda}{\lambda}\right)^{\alpha+1} \cdot \frac{1}{\lambda^2} = \boxed{\frac{\alpha(\alpha+1)}{\lambda^2}}$$

4.4.5 Variance

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{\alpha(\alpha+1)}{\lambda^2} - \left(\frac{\alpha}{\lambda}\right)^2 = \frac{\alpha}{\lambda^2}$$

$$\boxed{\mathrm{Var}(X) = \frac{\alpha}{\lambda^2}}$$

4.4.6 Questions

(a) Let $X \sim \text{Gamma}(\alpha = 3, \lambda = 2)$. Find the mean and variance of X.

(b) Show that if $X \sim \text{Exponential}(\lambda)$, then $X \sim \text{Gamma}(\alpha = 1, \lambda)$.

(c) The length of time for one individual to be served at a cafeteria is exponentially distributed with mean 4 minutes. Find the probability that the total service time for 6 individuals is less than 15 minutes.

(d) Suppose $X \sim \text{Gamma}(\alpha = 4, \lambda = 1)$. Calculate $P(X > 3)$.

(e) The sum of n independent Exponential(λ) random variables follows which distribution? Explain briefly.

(f) If $X \sim \text{Gamma}(\alpha = 2, \lambda = 3)$, compute $E[X^2]$ using the properties of the Gamma distribution.

(g) Suppose $X \sim \text{Gamma}(\alpha = 5, \lambda)$ and the mean of X is 10. Find λ.

(h) In a certain city, the daily consumption of electric power, in millions of kilowatt-hours, is a random variable X having a gamma distribution with mean $\mu = 6$ and variance $\sigma^2 = 12$. Find the values of the parameters α (shape) and λ (scale). Then Find the probability that on any given day the daily power consumption will exceed 12 million kilowatt-hours, i.e., find $P(X > 12)$.

4.5 The Normal Distribution

Let $X \sim N(\mu, \sigma^2)$ with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 > 0$.

The probability density function (pdf) is:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}$$

4.5.1 Proof that the PDF is Valid

We need to verify:

$$\int_{-\infty}^{\infty} f(x)\, dx = 1$$

Make substitution $z = \frac{x-\mu}{\sigma} \implies x = \sigma z + \mu$, so $dx = \sigma dz$.

Then

$$\int_{-\infty}^{\infty} f(x)\, dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}\, dz$$

Consider the integral without the normalizing constant:

$$I = \int_{-\infty}^{\infty} e^{-\frac{z^2}{2}}\, dz$$

We want to compute I. Since the integrand is positive and symmetric, define

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-\frac{x^2}{2}}\, dx\right)\left(\int_{-\infty}^{\infty} e^{-\frac{y^2}{2}}\, dy\right) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\frac{x^2+y^2}{2}}\, dx\, dy$$

Interpreting this as a double integral over $\mathbb{R}^2$:

$$I^2 = \iint_{\mathbb{R}^2} e^{-\frac{r^2}{2}}\,dx\,dy$$

where $r^2 = x^2 + y^2$.
Switch to polar coordinates:

$$x = r\cos\theta, \quad y = r\sin\theta, \quad dx\,dy = r\,dr\,d\theta$$

with $r \in [0, \infty)$, $\theta \in [0, 2\pi)$.
Thus,

$$I^2 = \int_0^{2\pi}\int_0^{\infty} e^{-\frac{r^2}{2}} r\,dr\,d\theta = \left(\int_0^{2\pi} d\theta\right)\left(\int_0^{\infty} re^{-\frac{r^2}{2}}\,dr\right) = 2\pi\int_0^{\infty} re^{-\frac{r^2}{2}}\,dr$$

Make substitution:

$$u = \frac{r^2}{2} \implies du = r\,dr$$

So,

$$\int_0^{\infty} re^{-\frac{r^2}{2}}\,dr = \int_0^{\infty} e^{-u}du = \left[-e^{-u}\right]_0^{\infty} = 1$$

Therefore,

$$I^2 = 2\pi \times 1 = 2\pi$$

Taking the positive root:

$$I = \sqrt{2\pi}$$

Finally, the normalized integral is

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}\,dz = \frac{1}{\sqrt{2\pi}} \times I = \frac{1}{\sqrt{2\pi}} \times \sqrt{2\pi} = 1$$

Hence,

$$\boxed{\int_{-\infty}^{\infty} f(x)dx = 1}$$

4.5.2 Moment Generating Function (MGF)

By definition,

$$M_X(t) = \mathbb{E}[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} f(x)dx = \int_{-\infty}^{\infty} e^{tx}\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)dx$$

Rewrite the exponent as

$$tx - \frac{(x-\mu)^2}{2\sigma^2} = -\frac{1}{2\sigma^2}\left((x-\mu)^2 - 2\sigma^2 tx\right)$$

Complete the square inside the parenthesis:

$$(x-\mu)^2 - 2\sigma^2 tx = (x-\mu)^2 - 2\sigma^2 t(x-\mu+\mu) = (x-\mu)^2 - 2\sigma^2 t(x-\mu) - 2\sigma^2 t\mu$$

Rewrite as:

$$(x-\mu)^2 - 2\sigma^2 t(x-\mu) = (x-\mu-\sigma^2 t)^2 - \sigma^4 t^2$$

Hence,

$$tx - \frac{(x-\mu)^2}{2\sigma^2} = -\frac{1}{2\sigma^2}\left[(x-\mu-\sigma^2 t)^2 - \sigma^4 t^2\right] - t\mu = -\frac{(x-\mu-\sigma^2 t)^2}{2\sigma^2} + \frac{\sigma^2 t^2}{2} + t\mu$$

Therefore,

$$M_X(t) = e^{t\mu + \frac{\sigma^2 t^2}{2}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu-\sigma^2 t)^2}{2\sigma^2}\right) dx$$

The integral is the integral of a normal PDF with mean $\mu + \sigma^2 t$ and variance σ^2, which integrates to 1. So:

$$\boxed{M_X(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)}$$

—

4.5.3 Moments from the MGF

The n-th moment is given by the n-th derivative of $M_X(t)$ evaluated at $t = 0$:

$$\mathbb{E}[X] = M_X'(0), \quad \mathbb{E}[X^2] = M_X''(0)$$

Calculate the first derivative:

$$M_X'(t) = \frac{d}{dt}\left(e^{\mu t + \frac{\sigma^2 t^2}{2}}\right) = e^{\mu t + \frac{\sigma^2 t^2}{2}}\left(\mu + \sigma^2 t\right)$$

At $t = 0$:

$$\mathbb{E}[X] = M_X'(0) = e^0 \cdot \mu = \boxed{\mu}$$

Calculate the second derivative:

$$M_X''(t) = \frac{d}{dt}\left[e^{\mu t + \frac{\sigma^2 t^2}{2}}(\mu + \sigma^2 t)\right]$$

Using product rule:

$$M_X''(t) = e^{\mu t + \frac{\sigma^2 t^2}{2}}(\mu + \sigma^2 t)^2 + e^{\mu t + \frac{\sigma^2 t^2}{2}}\sigma^2 = e^{\mu t + \frac{\sigma^2 t^2}{2}}\left[(\mu + \sigma^2 t)^2 + \sigma^2\right]$$

At $t = 0$:

$$\mathbb{E}[X^2] = M_X''(0) = e^0\left(\mu^2 + \sigma^2\right) = \boxed{\mu^2 + \sigma^2}$$

—

4.5.4 Variance

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = (\mu^2 + \sigma^2) - \mu^2 = \boxed{\sigma^2}$$

4.5.5 Questions

(a) Let $X \sim N(\mu = 50, \sigma = 5)$. Find $P(45 < X < 55)$.

(b) Suppose the heights of adult men in a city are normally distributed with mean 175 cm and standard deviation 8 cm. What percentage of men are taller than 185 cm?

(c) If $X \sim N(100, 15^2)$, calculate $P(X < 85)$.

(d) A machine produces rods with lengths normally distributed with mean 20 cm and variance 4 cm^2. Find the probability that a randomly selected rod has length between 18 cm and 22 cm.

(e) The scores on a certain exam are normally distributed with mean 70 and standard deviation 10. Find the cutoff score for the top 10% of the students.

(f) If $X \sim N(0, 1)$, find $P(|X| > 2)$.

(g) The average daily sales (in thousands) of a store are normally distributed with mean 12 and standard deviation 3. What is the probability that on a given day sales exceed 18 thousand?

(h) Suppose the weight of apples in a shipment is normally distributed with mean 150 grams and standard deviation 20 grams. What is the probability an apple weighs less than 130 grams?

(i) A blood pressure measurement is normally distributed with mean 120 mmHg and standard deviation 15 mmHg. Find the probability that a measurement lies between 110 and 140 mmHg.

(j) A company packs sugar into 1 kg bags. The weight is normally distributed with mean 1 kg and standard deviation 0.02 kg. What percentage of bags weigh less than 0.97 kg?

5 Central Limit Theorem (CLT)

Statement:

Let $X_1, X_2, \ldots, X_n$ be a random sample of size n drawn from any population with mean μ and finite variance σ^2. Then, as $n \to \infty$, the sampling distribution of the sample mean $\bar{X}$ approaches a normal distribution:

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

Equivalently, the standardized variable

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1) \quad \text{as } n \to \infty.$$

5.1 Key Points

- The original population distribution does not need to be normal.
- The population must have a finite mean and variance.
- As the sample size n increases, the distribution of $\bar{X}$ becomes approximately normal.
- A commonly used rule: the normal approximation is reasonable when $n \geq 30$.

5.2 Applications

- Constructing confidence intervals for the population mean.
- Performing hypothesis testing about the population mean.
- Justifying the use of normal models in sampling-based inference even for non-normal populations.

5.3 Questions

(a) A population has mean $\mu = 80$ and standard deviation $\sigma = 12$. If a sample of size $n = 36$ is taken, find the probability that the sample mean $\bar{X}$ is less than 78.

(b) The weights of sugar packets are normally distributed with mean 1 kg and standard deviation 0.05 kg. Find the probability that the mean weight of a sample of 25 packets lies between 0.99 kg and 1.01 kg.

(c) A machine fills soda bottles with a mean volume of 500 ml and standard deviation of 5 ml. If a random sample of 40 bottles is selected, what is the probability that their average volume exceeds 502 ml?

(d) The average time a student takes to complete an online exam is 90 minutes with a standard deviation of 15 minutes. A sample of 64 students is selected. What is the probability that their average completion time is more than 93 minutes?

(e) A factory produces metal rods whose lengths are skewed right with a mean of 100 cm and a standard deviation of 20 cm. If a sample of 49 rods is selected, what is the probability that their average length is between 98 cm and 102 cm?

(f) A population has unknown distribution but with mean $\mu = 250$ and standard deviation $\sigma = 40$. A sample of size $n = 100$ is taken.

(a) What is the standard error of the sample mean?

(b) What is the probability that the sample mean lies within 5 units of the population mean?

(g) A survey finds that the mean daily screen time among teenagers is 6 hours with a standard deviation of 2 hours. What is the minimum sample size required so that the probability that the sample mean differs from the population mean by less than 0.5 hours is at least 0.95?

(h) The life expectancy of a certain species of plants is normally distributed with mean 150 days and standard deviation 30 days. If a random sample of 25 plants is selected, what is the probability that their average life expectancy is between 140 and 160 days?

6 Point Estimation

6.1 Properties of Good Estimator

6.1.1 Unbiasedness & Sufficiency

A network of sensors is deployed to monitor the ambient temperature in a greenhouse. Each sensor records temperature readings independently. The readings from four sensors are modeled as i.i.d. random variables X_1, X_2, X_3, X_4, where each X_i is an unbiased measurement of the true average temperature μ, i.e., $E[X_i] = \mu$, and $\text{Var}(X_i) = \sigma^2$.

Two estimators are proposed for estimating the average temperature μ:

$$\hat{\mu}_A = \frac{X_1 + X_2 + X_3 + X_4}{4}, \qquad \hat{\mu}_B = 0.6X_1 + 0.1X_2 + 0.2X_3 + 0.1X_4$$

(a) Are $\hat{\mu}_A$ and $\hat{\mu}_B$ unbiased estimators of the true temperature μ? Show your reasoning.

(b) Compute the variances of $\hat{\mu}_A$ and $\hat{\mu}_B$ in terms of σ^2.

(c) Which estimator is more efficient? Justify your answer based on the variances.

6.1.2 Sufficiency

(a) Define what it means for a statistic to be **sufficient** for a parameter. Explain in your own words.

(b) Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with an unknown parameter θ. Explain why the sample mean $\bar{X}$ might be considered a sufficient statistic for θ in some cases.

(c) Suppose you are given a random sample $X_1, X_2, \ldots, X_n$ from a distribution with unknown parameter θ, and a statistic $T(X)$, which is some function of the sample (e.g., the sample mean or the sum of the sample values). How would you determine whether $T(X)$ is a sufficient statistic for θ? Describe the approach.

6.1.3 Consistency

(a) What does it mean for an estimator $\hat{\theta}_n$ to be **consistent** for a parameter θ? Describe in simple terms.

(b) Suppose the sample mean $\bar{X}$ is used to estimate the population mean μ. Explain why $\bar{X}$ is a consistent estimator of μ.

(c) In your own words, explain what happens to a consistent estimator as the sample size n becomes very large.

6.2 Point Estimation: Maximum Liklihood Estimator(MLE)

Question The Weibull distribution is commonly used to model time-to-failure data in reliability engineering and survival analysis. The probability density function (PDF) of the Weibull distribution is given by:

$$f(x; \lambda, k) = \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k}, \quad x \geq 0$$

where $\lambda > 0$ is the scale parameter and $k > 0$ is the shape parameter.

Suppose you observe a random sample $x_1, x_2, \ldots, x_n$ from a Weibull distribution with known shape parameter k and unknown scale parameter λ.

(a) Write down the likelihood function $L(\lambda)$ based on the sample.

(b) Derive the log-likelihood function $\ell(\lambda)$.

(c) Find the maximum likelihood estimator (MLE) $\hat{\lambda}$ of the scale parameter λ.

(d) Suppose $n = 4$, $k = 2$, and the observed data are:

$$x = \{1.5, 2.0, 2.3, 1.7\}$$

Compute the MLE $\hat{\lambda}$ for this dataset.

6.3 Confidence Interval: Mean and Difference of Means

6.3.1 Confidence Interval for the Mean (Known Population Standard Deviation)

Formula

$$\text{CI} = \bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

Derivation
We start from the standard normal distribution:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

For a confidence level of $1-\alpha$:

$$P\left(-z_{\alpha/2} \leq \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right) = 1-\alpha$$

Multiplying through by $\frac{\sigma}{\sqrt{n}}$:

$$P\left(\bar{X} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}\right) = 1-\alpha$$

Therefore, the confidence interval is:

$$\boxed{\bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}}$$

6.3.2 Confidence Interval for the Mean (Unknown σ, Large Sample)

Formula

$$\text{CI} = \bar{x} \pm z_{\alpha/2} \cdot \frac{S}{\sqrt{n}}$$

Justification:

When the sample size $n \geq 30$, the sample standard deviation s can be used in place of the population standard deviation σ by the Central Limit Theorem. The sampling distribution of the mean is approximately normal:

$$\frac{\bar{X}-\mu}{S/\sqrt{n}} \approx N(0,1)$$

Thus, the confidence interval is approximately:

$$P\left(\bar{X} - z_{\alpha/2} \cdot \frac{S}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2} \cdot \frac{S}{\sqrt{n}}\right) \approx 1-\alpha$$

Final form:

$$\boxed{\bar{x} \pm z_{\alpha/2} \cdot \frac{S}{\sqrt{n}}}$$

Confidence Interval for the difference of Means (known σ_1 and σ_2)

$$\boxed{(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \cdot \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

6.3.3 Confidence Interval for the difference of Means (Unknown σ_1 and σ_2, Large Samples)

$$\boxed{(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \cdot \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

6.3.4 Questions:

(a) A study of 120 hospitals reports a mean of 200 ER patients per day, with population standard deviation 25. Construct a 99.5% confidence interval.

(b) A sample of 75 laptops has a mean battery life of 6.8 hours. The population standard deviation is 1.2 hours. Construct a 97% confidence interval.

(c) A marketing firm estimates the average phone screen time. In a sample of 150 people, the mean is 4.3 hours with a population standard deviation of 0.8 hours. Find both 90% and 99% confidence intervals and discuss how the level of confidence affects interval width.

(d) A company claims the average data analyst salary is \$80,000. A sample of 100 shows a mean of \$78,500 with a population standard deviation of \$5,000. Construct a 95% confidence interval and evaluate the company's claim.

(e) A researcher conducts a hypothesis test at significance level $\alpha = 0.05$. (a) Define Type I error and Type II error in this context. (b) What does increasing α do to the probabilities of Type I and Type II errors?

(f) Suppose a test is performed to detect if a new drug reduces blood pressure. The null hypothesis is that the mean reduction is zero. Given a fixed sample size and standard deviation, how does increasing the true mean reduction affect the power of the test?

(g) Consider a 95% confidence interval for a population mean with known standard deviation. Explain how the width of the confidence interval changes when: (a) The sample size increases. (b) The significance level α increases from 0.05 to 0.10. (c) The population standard deviation increases.

(h) A quality control manager wants to test if the average diameter of manufactured parts differs from the target diameter of 50 mm. The test is conducted at $\alpha = 0.01$ with a sample size of 30 and standard deviation of 2 mm. If the sample mean shifts by 1 mm, discuss how this affects the power of the test. Also, describe how changing α or sample size would affect the confidence interval for the mean diameter.

6.3.5 Questions: Confidence Interval for Difference of Two Means

(a) Sample A: $n_1 = 100$, $\bar{x}_1 = 65$, $\sigma_1 = 10$
Sample B: $n_2 = 120$, $\bar{x}_2 = 60$, $\sigma_2 = 12$
Construct a 95% confidence interval for the difference in means $\mu_1 - \mu_2$.

(b) A company tests two production lines. Line A ($n = 80$) has an average output time of 25 mins, $\sigma = 2.4$ mins. Line B ($n = 100$) has mean time 26.1 mins, $\sigma = 2.8$ mins. Construct a 90% confidence interval for the difference in mean times.

(c) Two types of fertilizers are tested on crop yield. Fertilizer A ($n = 60$, $\bar{x} = 2400$ kg/acre, $\sigma = 120$), Fertilizer B ($n = 50$, $\bar{x} = 2250$ kg/acre, $\sigma = 130$). Find a 99% confidence interval for the difference in mean yields.

(d) A marketing team measures average monthly spending for two customer segments:
Segment 1: $n = 150$, mean = \$220, $\sigma = 30$
Segment 2: $n = 140$, mean = \$210, $\sigma = 28$
Construct a 95% confidence interval for the difference in average spending.

7 Hypothesis Testing: Mean and Difference of Means

7.1 Testing a Single Mean

Q1. A sample of 50 students has an average score of 78 with a known population standard deviation of 10. Test whether the population mean is 75 at $\alpha = 0.05$.

Q2. A tech recruiter claims that applicants for a new programming bootcamp score lower than 75 on a coding aptitude test. From a random sample of 36 applicants, the average score is 73 with a known population standard deviation of 6.At the 0.05 level of significance, perform the hypothesis test. Compute the p-value and interpret the result.

Q3. A manufacturer claims the average lifetime of a battery is 100 hours. A random sample of 32 batteries has a mean life of 96 hours and sample standard deviation of 8 hours. Test the claim at the 0.01 significance level (assume normality).

Q4. A company claims its machines produce rods with a mean length of 10 cm. A quality control engineer takes a sample of 30 rods, finds a mean of 9.8 cm, and a sample standard deviation of 0.4 cm. Test the company's claim at $\alpha = 0.05$.

Q5. A software company claims that its new algorithm improves processing speed to more than 120 milliseconds per request. A sample of 50 requests is taken, with a mean time of 122 ms and a population standard deviation of 5 ms.At the 0.01 significance level, test the claim using a z-test. Interpret the result.

Q6. A factory produces light bulbs with lifetimes normally distributed with mean 1000 hours and standard deviation 100 hours. A quality control test is set up with:

$$H_0 : \mu = 1000, \quad H_a : \mu < 1000,$$

at significance level $\alpha = 0.05$. The test rejects H_0 if the sample mean lifetime $\bar{X}$ from a sample of size 36 is less than 970 hours.

(a) Calculate the probability of a Type I error.

(b) Calculate the probability of a Type II error if the true mean lifetime is 950 hours.

Q7. For the same setup in Question 1, find the power of the test when the true mean lifetime is 950 hours.

Q8. A hypothesis test is performed to check if a new teaching method improves test scores. The null hypothesis is $H_0 : \mu = 75$ and the alternative is $H_a : \mu > 75$. Suppose the test uses $\alpha = 0.01$, sample size 25, and population standard deviation $\sigma = 10$. If the true mean score is actually 80, calculate the Type II error β and power of the test.

Q9. Explain how the width of a confidence interval for a population mean is affected by:

(a) Increasing the confidence level (e.g., from 95% to 99%)

(b) Increasing the sample size

(c) Increasing the population standard deviation

Q10. A pharmaceutical company tests if a drug decreases cholesterol levels. The test uses $\alpha = 0.05$ and sample size 40 with $\sigma = 15$. The null hypothesis is $H_0 : \mu = 200$, alternative $H_a : \mu < 200$. Calculate the Type II error if the true mean cholesterol level is 185, and discuss how increasing the sample size affects the Type II error.

Q11. Discuss the trade-off between Type I and Type II errors. How does changing the significance level α affect these errors and the power of a test?

7.2 Testing the Difference of Two Independent Means

(a) The mean weight of widgets from machine A is 50g (sample size 40, SD = 5g), and from machine B is 52g (sample size 35, SD = 6g). Test whether there is a significant difference in mean weights at $\alpha = 0.05$.

(b) Two teaching methods are tested on two independent groups. Group 1 (n=33) has a mean score of 85 (SD = 6), and Group 2 (n=31) has a mean score of 82 (SD = 5). Test whether the first method leads to higher scores using a one-tailed test at 1% level of significance.

(c) Two data centers report different average response times. A sample of 100 requests from Data Center A yields a mean response time of 210 ms with a population standard deviation of 15 ms. A sample of 120 requests from Data Center B yields a mean of 215 ms with a population standard deviation of 20 ms. Test at the 0.05 significance level whether there is a significant difference in mean response times.

7.3 Hypothesis Testing: Regression Parameters

7.3.1 Simple Linear Regression (Large Sample)

Q1. In the simple linear regression model $y = \beta_0 + \beta_1 x + \varepsilon$, define the null and alternative hypotheses for testing the slope β_1 using a z-test.

Q2. A simple regression analysis yields:

$$\hat{y} = 1.2 + 3.4x, \quad SE_{\beta_1} = 0.5$$

Assume $n = 1000$ (a large sample). Test whether β_1 is significantly different from zero at the 5% significance level using the z-test.

7.3.2 Multiple Linear Regression (Large Sample)

(a) For the multiple regression model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon$$

state the null and alternative hypotheses to test if $\beta_2 = 0$ using a z-test.

(b) A multiple regression model output shows:

$$\hat{\beta}_2 = 0.75, \quad SE_{\beta_2} = 0.25, \quad n = 1200$$

Use a z-test to determine whether x_2 significantly affects y at the 1% significance level.

(c) A digital marketing team develops a multiple linear regression model to predict the weekly number of purchases (y) from an e-commerce website using the following inputs:

- x_1: Weekly advertising budget on social media (in $1000s)
- x_2: Website traffic (in 1000s of visits)
- x_3: Average session duration (in minutes)

The estimated regression equation is:

$$\hat{y} = 10 + 1.8x_1 + 2.5x_2 + 0.1x_3$$

The standard errors of the estimated coefficients based on a sample size of $n = 1500$ are:

$$SE_{\beta_1} = 0.4, \quad SE_{\beta_2} = 0.6, \quad SE_{\beta_3} = 0.3$$

Test whether each predictor significantly affects the weekly number of purchases at the 1% significance level using a z-test.

8 Logistic Regression

8.1 Sigmoid Function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

8.1.1 Symmetry:

Show that $\sigma(-z) = 1 - \sigma(z)$

$$\sigma(-z) = \frac{1}{1+e^{z}}, \quad \sigma(z) = \frac{1}{1+e^{-z}}$$

$$1 - \sigma(z) = 1 - \frac{1}{1+e^{-z}} = \frac{e^{-z}}{1+e^{-z}} = \frac{1}{1+e^{z}} = \sigma(-z)$$

8.1.2 Inverse:

Solve $p = \sigma(z)$ for z

$$p = \frac{1}{1+e^{-z}} \Rightarrow 1 + e^{-z} = \frac{1}{p} \Rightarrow e^{-z} = \frac{1-p}{p}$$

$$\Rightarrow -z = \log\left(\frac{1-p}{p}\right) \Rightarrow z = \log\left(\frac{p}{1-p}\right)$$

8.1.3 Derivative:

$$\sigma(z) = \frac{1}{1+e^{-z}} \Rightarrow \frac{d\sigma}{dz} = \frac{e^{-z}}{(1+e^{-z})^2}$$

$$= \left(\frac{1}{1+e^{-z}}\right)\left(1 - \frac{1}{1+e^{-z}}\right) = \sigma(z)(1-\sigma(z))$$

8.2 Introduction to Logistic Regression

Purpose: Logistic regression is used to model the probability of a binary outcome (e.g., success/failure, yes/no, 0/1) based on one or more predictor variables.

Model Form:

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \cdots + \beta_j X_j)}}$$

Log-Odds (Logit) Form:

$$\log\left(\frac{P(Y=1)}{1-P(Y=1)}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_j X_j$$

Why Use It?

- Predicts probability while keeping values in [0, 1]
- Interpretable via odds and log-odds
- Suitable for classification problems

8.2.1 Why Logistic Regression Has This Form

Problem: We want to model a probability $P(Y = 1 \mid X = x)$, but a linear model like $\beta_0 + \beta_1 x$ can take values outside the interval [0, 1].

Solution: Use the *logit function*, which maps probabilities to the entire real line:

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right)$$

Model:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x \quad \Rightarrow \quad p = \frac{1}{1 + e^{-(\beta_0+\beta_1 x)}}$$

Why this form?

- Ensures $0 < p < 1$ for all x
- Logit is the inverse of the sigmoid function
- Interpretable: coefficients show how log-odds change with x

Meaning of Logistic Coefficient β_j **Model:**

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_j X_j$$

Interpretation:

- Each β_j represents the change in **log-odds** of $Y = 1$ for a one-unit increase in X_j.
- Exponentiating β_j gives the **odds ratio**:

$$\text{Odds Ratio} = e^{\beta_j}$$

- If $\beta_j > 0$, the odds of $Y = 1$ increase with X_j.
- If $\beta_j < 0$, the odds decrease.

8.2.2 Decision Boundary in Logistic Regression

$$P(Y = 1 \mid \mathbf{X}) = \frac{1}{1 + e^{-(\beta_0+\beta_1 X_1+\cdots+\beta_j X_j)}}$$

Decision Boundary: Predicted class changes at the threshold where

$$P(Y = 1 \mid \mathbf{X}) = 0.5$$

Start with:

$$\frac{1}{1 + e^{-(\beta_0+\beta_1 X_1+\cdots+\beta_j X_j)}} = 0.5$$

Solve:

$$1 + e^{-(\beta_0+\beta_1 X_1+\cdots+\beta_j X_j)} = 2$$

$$e^{-(\beta_0+\dots)} = 1 \quad \Rightarrow \quad \beta_0 + \beta_1 X_1 + \cdots + \beta_j X_j = 0$$

Hence Decision Boundary:

$$\beta_0 + \beta_1 X_1 + \cdots + \beta_j X_j = 0$$

- It is a **hyperplane** in the input space.

- Separates predictions where $P(Y = 1) > 0.5$ from those where $P(Y = 1) < 0.5$.
- Equivalent to a linear classifier boundary.

Questions

(a) A logistic regression model outputs a probability of 0.87 that a given email is spam. The threshold to classify an email as spam is 0.5. Should the email be classified as spam or not spam? Justify your answer.

(b) A logistic regression model outputs that the odds of a user clicking on an ad are 7:3. What is the corresponding probability?

(c) A logistic regression model predicts whether a user will click on a recommended item ($y = 1$), based on the item popularity score (x_1) and user activity level (x_2). The model is:

$$P(y = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(-2+0.5x_1+0.1x_2)}}$$

Calculate the probability that a user with popularity score 6 and activity level 40 clicks on the recommendation.

(d) A logistic regression model is used to predict whether a user will buy a product ($y = 1$) or not ($y = 0$), based on the time they spent on the website (in minutes), denoted by x. The model is:

$$P(y = 1 \mid x) = \frac{1}{1 + e^{-(\beta_0+\beta_1 x)}}$$

Suppose the parameters of the model are:

$$\beta_0 = -2, \quad \beta_1 = 0.5$$

You are given the following training data:

x	1	2	3	4
y	0	0	1	1

Compute the log-likelihood function for this dataset under the given model.

(e) Derive the gradient (first derivative) of the log-likelihood function with respect to β_0 and β_1.

(f) A company is developing a spam detection system using logistic regression. The model predicts whether an email is spam ($y = 1$) or not spam ($y = 0$) based on a single feature x, which represents the number of spam-related keywords found in the email.

The logistic regression model is given by:

$$P(y = 1 \mid x) = \frac{1}{1 + e^{-(\beta_0+\beta_1 x)}}$$

with parameters:

$$\beta_0 = 0.2, \quad \beta_1 = 1.5$$

The following emails are analyzed, with the number of spam keywords found in each:

$$x = \{0, 1, 2\}$$

(a) Calculate the predicted probability $P(y = 1 \mid x)$ for each value of x.

(b) Classify each email as spam or not spam using a threshold of 0.7. If $P(y = 1) \geq 0.7$, classify as spam ($y = 1$); otherwise, classify as not spam ($y = 0$).

(g) A logistic regression model is used to predict whether a customer will purchase a product ($y = 1$) based on their browsing time x (in minutes). The model is given by:

$$P(y = 1 \mid x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

Browsing Time (x)	1.5	2.5	3.5	4.5
Purchase (y)	0	1	1	0

(a) Write the likelihood function $L(\beta_0, \beta_1)$ based on the given data.

(b) Compute the likelihood value when $\beta_0 = 0$ and $\beta_1 = 1$.